Amanda Juvera

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Education

University of Michigan Aug 2023 – May 2027
B.S. Computer Science *Ann Arbor, MI*

- Relevant Coursework: Operating Systems, Advanced Operating Systems Projects, Data Structures and Algorithms, Machine Learning, Web Systems (in progress)
- Awards: Blue Ribbon Undergraduate Research Award, University Honors, LSA Scholarship Recipient

Experience

Software Engineer Intern May 2026 – Aug 2026
Oracle *Santa Clara, CA*

- Redesigned Syft-based Software Bill of Materials (SBOM) generation for an internal security scanning platform to preserve package ownership context, eliminating 100% of false-positive vulnerability audits for SmartNIC artifacts and surfacing 1000+ previously missed vulnerabilities, letting security engineers trust the tool instead of triaging noise.
- Engineered a custom build service configuration to scale security scanning across all Oracle Cloud Infrastructure artifacts, extending coverage beyond the Virtual Cloud Networking Card Management team.

Software Engineer Intern May 2025 – Aug 2025
Retrospect *Ann Arbor, MI*

- Shipped traffic light detection end-to-end into RiskEngine, an in-vehicle monitoring tool used to assess autonomous vehicle safety.
- Built a computer vision pipeline using HSV filtering and ROI cropping to classify traffic light states from low-quality video at 90% accuracy, replacing manual annotation and saving reviewers 10+ hours/week.

Computer Science Mentor Aug 2024 – Present
University of Michigan, Computer Science and Engineering *Ann Arbor, MI*

- Mentor students in data structures, memory management, debugging, and C++ systems programming, translating dense concepts into explanations that meet each student where they are.

Research Assistant Oct 2023 – Apr 2024
University of Michigan, Maldonado Lab *Ann Arbor, MI*

- Built LabVIEW automation software for Scanning Electrochemical Probe Microscopy experiments, cutting manual experiment time by 50%.

Projects

Social Media Analytics Pipeline | *Python, Notion API, GitHub Actions, LiveDune API*

- Built a nightly automation ingesting third-party marketing data into a client's Notion workspace, modeling records as Notion databases and handling API pagination, schema mapping, and rate limits.
- Managed encrypted secrets and dual cron schedules for reliable unattended runs, saving a social media company 5+ hours/week of manual reporting.

Navi — Route-Aware Navigation App | *TypeScript, React, PostgreSQL/PostGIS, Mapbox* *In Progress*

- Building a navigation app around an "On The Way" search that ranks stops by real detour cost, buffering the route polyline into a corridor with PostGIS and computing added travel time per candidate instead of searching by radius.
- Designing a fuel-stop scheduler that uses tank capacity, MPG, and reserve threshold to recommend stops, with selectable strategies for fewest stops versus lowest total cost.

Operating System Componenets | *C++*

- **Network File Server:** concurrent, crash-consistent server over TCP sockets, using fine-grained reader-writer locking with hand-over-hand traversal and ordered disk writes to preserve metadata invariants.
- **Virtual Memory Manager:** demand-paged system with swap- and file-backed pages, copy-on-write `fork()`, and clock page replacement to minimize disk I/O.
- **Thread Library:** preemptive kernel-level threading across multiple CPUs with `ucontext` switching, mutexes, condition variables, and interrupt-safe synchronization.

Skills

Languages: Python, C++, C, Java
Web & APIs: React, Node.js, REST APIs, Notion API, GitHub Actions, CI/CD
Systems & Tools: Linux, multithreading, POSIX threads/sockets, virtual memory, Git, GDB, LLDB, OCI, LabVIEW
Interests: Drawing, reading, hiking, weight-lifting